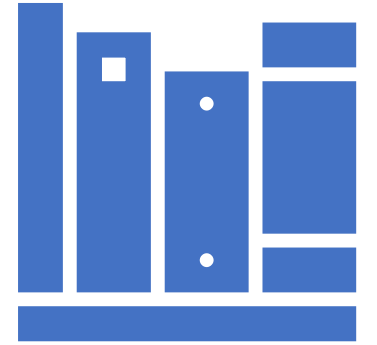




Module IV- Syntax-Driven Semantic Analysis

Dr.S.Periyanayagi,
Associate Professor/SCSE

Syntax Driven Semantic Analysis

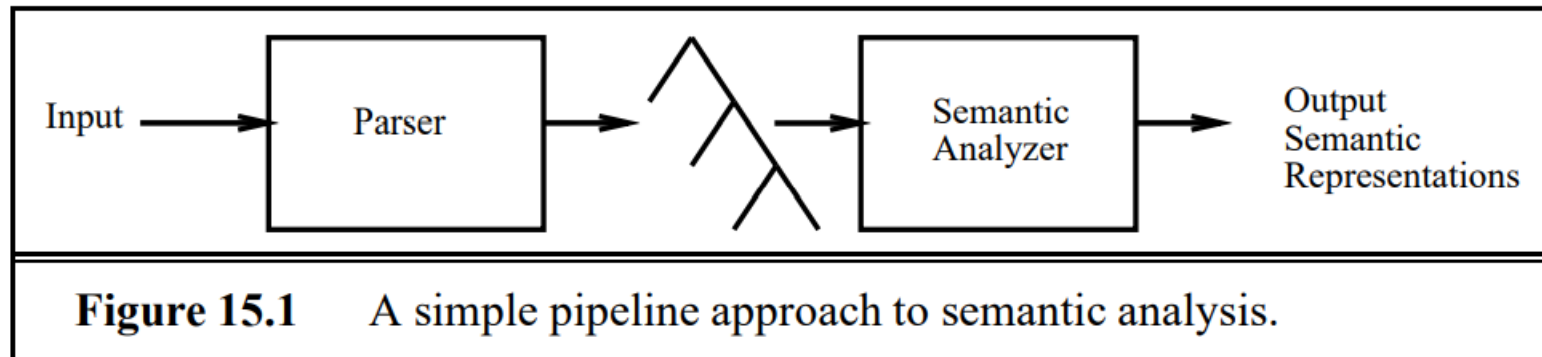


- Number of computational approaches to the problem of semantic analysis
- Syntax-driven semantic analysis – First Approach
- Based on **Principle of compositionality** - The key idea is that the **meaning of a sentence** can be composed from the **meanings of its parts**.
- The meaning of a sentence - not based solely on the words that make it up.
- It is based on the **ordering, grouping, and relations among the words** in the sentence

Contd...

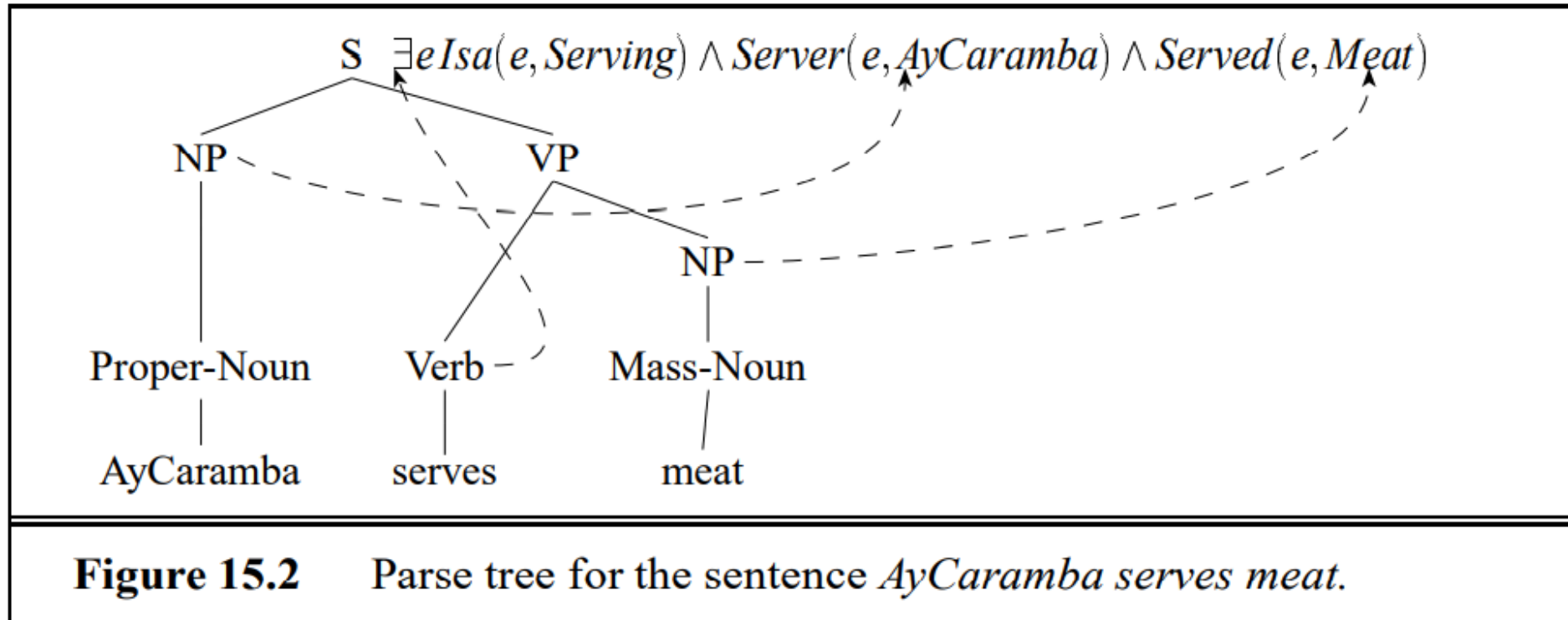


- Syntax-driven semantic analysis, the composition of meaning representations is guided by the syntactic components and relations.
- An input is first passed through a parser to derive its syntactic analysis.



- This analysis is then passed as input to a semantic analyzer to produce a meaning representation

Parse Tree





Contd...

- The grammar rules and the lexical entries. This is known as the rule-to-rule hypothesis.
- Designing an analyzer based on this approach brings us back to the notion of parts and what it means for them to have meanings.



Semantic Augmentations to CFG Rules

- Augmenting context-free grammar rules with semantic attachments
- Abstractly, our augmented rules have the following structure

$$A \rightarrow \alpha_1 \dots \alpha_n \quad \{f(\alpha_j.sem, \dots, \alpha_k.sem)\}$$

- The semantic attachment to the basic context-free rule is shown in the $\{ \dots \}$ to the right of the rule's syntactic constituents.
- This notation states that the meaning representation assigned to the construction A , which we will denote as $A.sem$, can be computed by running the function f on some subset of the semantic attachments of A 's constituents.

Contd...



- The concrete entities in this example are represented by the FOPC constants *AyCaramba* and *Meat*.
- Our first task is to associate these constants with the constituents of the tree that introduce them.
- The first step toward accomplishing this is to pair them with the lexical rules representing the words that introduce them into the sentence.

ProperNoun $\rightarrow$ *AyCaramba* $\{AyCaramba\}$

MassNoun $\rightarrow$ *meat* $\{Meat\}$

- These two rules specify that the meanings associated with the subtrees generated by these rules.

Contd...



- The subtrees corresponding to these rules do not directly contribute these FOPC constants to the final meaning representation.
- It is the NPs higher in the tree that contribute them to the final representation.
- In keeping with the principle of compositionality, we can deal with this indirect contribution by stipulating that the upper NPs obtain their meaning representations from the meanings of their children.
- In these two cases, the meaning representations of the children are simply copied upward to

NP – Proper Noun {ProperNoun.sem}

NP - MassNoun {MassNoun.sem}



Contd...

- These rules state that the meaning representation of the noun phrases are the same as the meaning representations of their individual components, denoted by *ProperNoun.sem* and *MassNoun.sem*.
- Non-branching grammar rules, the semantic expression associated with the child will be copied unchanged to the parent.
- While the static semantic attachment to our first NP rule is simply *ProperNoun.sem*, the semantic value of the tree produced by that rule in this example is *AyCaramba*.

Contd...



- A generic Serving event involves a Server and something Served, as captured in the following logical formula

$$\exists e, x, y \text{ Isa}(e, \text{Serving}) \wedge \text{Server}(e, x) \wedge \text{Served}(e, y)$$

- As a first attempt at this verb's semantic attachment, we can simply take this logical formula as serve's semantic attachment

$$\text{Verb} \rightarrow \text{serves}$$

$$\{\exists e, x, y \text{ Isa}(e, \text{Serving}) \wedge \text{Server}(e, x) \wedge \text{Served}(e, y)\}$$

Contd...



- Moving up the parse tree, the next constituent to be considered is the VP that dominates both serves and meat.

$$\exists e, x \text{ Isa}(e, \text{Serving}) \wedge \text{Server}(e, x) \wedge \text{Served}(e, \text{Meat})$$

- Unlike the NPs, we can not simply copy the meaning of these children up to the parent VP.
- we need to incorporate the meaning of the NP into the meaning of the Verb and assign the resulting representation to the VP.sem.
- This consists of replacing the **variable y** with the **logical term Meat** as the second argument of the Served role of the Serves event.

Contd..



- The semantic attachment for the VP must provide a means to replace the quantified variable y within the body of $V.sem$ with the logical constant $Meat$, as stipulated by $NP.sem$.

$$\exists e, x \text{ Isa}(e, \text{Serving}) \wedge \text{Server}(e, x) \wedge \text{Served}(e, \text{Meat})$$

- Verb's semantic attachment are to be replaced by the semantics of the Verb's arguments, and the ability to perform such a replacement — No Straight forward way.



Lambda Notation

- A notational extension to FOPC called the **lambda notation** that provides replacement.
- This notation extends the syntax of FOPC to include expressions of the following form.

$$\lambda xP(x)$$

- Such expressions consist of the **Greek symbol λ** , followed by one or more variables, followed by a FOPC expression.
- λ -expressions is based on the ability to apply them to logical terms to yield new FOPC expressions where the formal parameter variables are bound to the specified terms.

Contd...

- This process is known as λ -reduction - accompanied by the subsequent removal of the λ .
- Expressions - the application of a λ -expression to the constant A, followed by the result of performing a λ reduction on this expression.

$$\lambda x P(x)(A)$$
$$P(A)$$

- This λ -notation provides both of the capabilities in the Verb semantics and the λ -reduction process implements the desired replacement of variables with terms.



Contd...



- An important and useful variation of this technique is the use of one λ -expression as the body of another as in the following expression.

$$\lambda x \lambda y \text{ Near}(x; y)$$

λ -expression - $\lambda x \lambda y \text{ Near}(x; y)(\text{ICSI})$

λ -removal - $\lambda y \text{ Near}(\text{ICSI} ; y)$

This resulting λ -expression can, in turn, be applied to another term to arrive at a fully specified logical formula.

$$\lambda y \text{ Near}(\text{ICSI} ; y)(\text{AyCaramba})$$
$$\text{Near}(\text{ICSI} ; \text{AyCaramba})$$



Contd...

- This technique, called **currying** is a way of converting a predicate with multiple arguments into a sequence of single argument predicates.
- This can be accomplished in two steps: **changing the semantic attachment of the Verb to a λ -expression**, and having the **semantic attachment of the VP apply this expression to the NP semantics**.
- The first of these steps can be accomplished by designating x , the variable corresponding to the Served role, as the λ -variable for a λ -expression provided as the semantic attachment for serve.

Verb $\rightarrow$ *serves*

$\{\lambda x \exists e, y \text{ Isa}(e, \text{Serving}) \wedge \text{Server}(e, y) \wedge \text{Served}(e, x)\}$



Contd...

- λ -application where the λ -expression is provided by *Verb.sem* and the argument is provided by *NP.sem*

$$VP \rightarrow Verb\ NP \quad \{Verb.sem(NP.sem)\}$$

- A λ -reduction removes the λ revealing the inner expression with the parameter *x* replaced by the constant *Meat*.
- This expression, the meaning of the verb phrase serves meat, is then the value of *VP.sem*

$$\exists e, y\ Isa(e, Serving) \wedge Server(e, y) \wedge Served(e, Meat)$$

Contd...



- S rule $S \rightarrow NP VP \quad \{VP.sem(NP.sem)\}$
- stands the value of VP.Sem doesn't provide the necessary λ expression.
- The lambda-application performed at the VP rule resulted in a generic FOPC expression with two existentially quantified variables.
- The Verb attachment should instead have consisted of an embedded λ -expression to make the Server role available for binding at the S level of the grammar.
- Revised representation of the Verb attachment will be the following.

Verb $\rightarrow$ *serves*

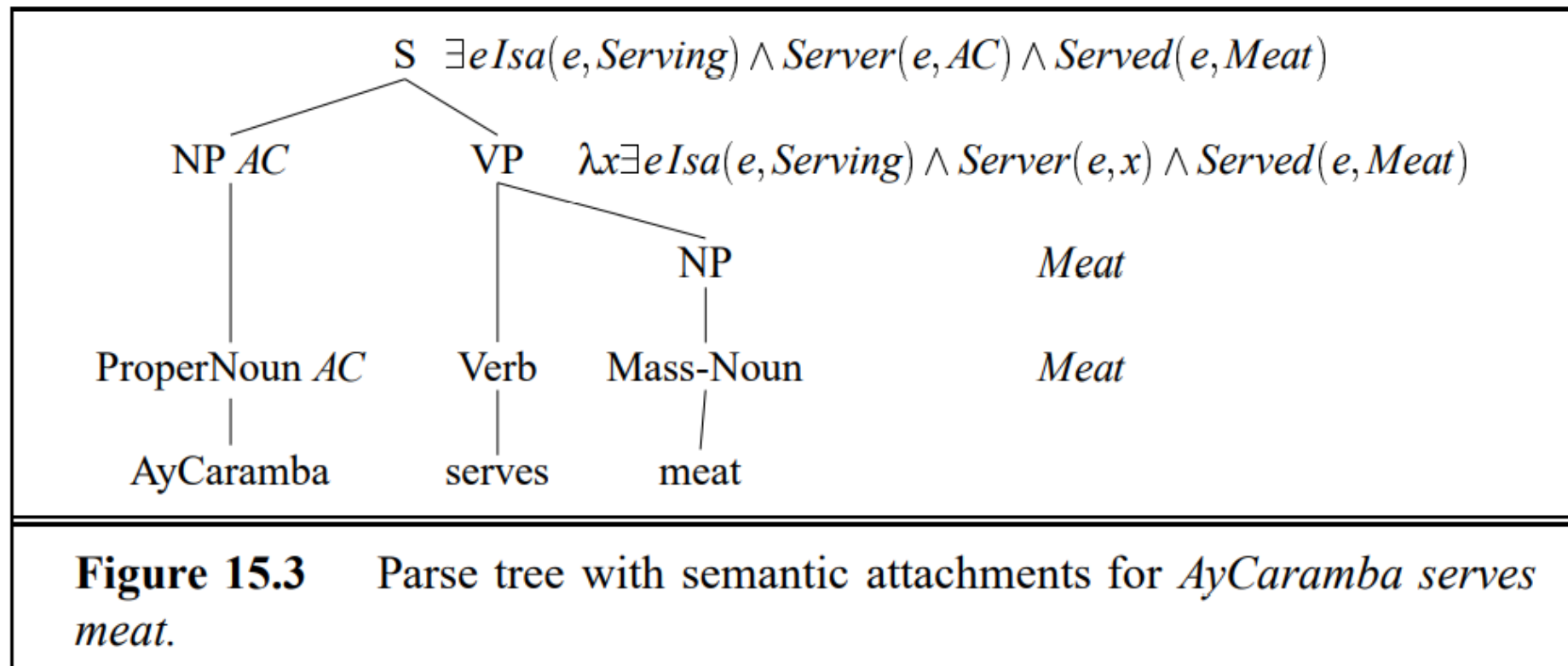
$\{\lambda x \lambda y \exists e Isa(e, Serving) \wedge Server(e, y) \wedge Served(e, x)\}$

Contd...



- The body of this Verb attachment consists of a λ -expression inside a λ -expression.
- The outer expression provides the variable that is replaced by the first λ -reduction, while the inner expression can be used to bind the final variable corresponding to the Server role.
- The parse tree for this example, with each node annotated with its corresponding semantic.

Contd...



Contd...



A restaurant serves meat.

- formula is a plausible representation for the meaning of the subject in this example

$$\exists x Isa(x, Restaurant)$$

- Combining this new representation with the one already developed for the verb phrase, yields the following meaning representation.

$$\begin{aligned} &\exists e, x Isa(e, Serving) \\ &\quad \wedge Server(e, x) \wedge Served(e, Meat) \wedge Isa(x, Restaurant) \end{aligned}$$

- In this formula, the restaurant, represented by the variable x , is specified as playing the role of the Server by its presence as the second argument to the Server predicate.

Contd...



- Unfortunately, the λ -application specified as the semantic attachment for the S rule will not produce this result.
- A literal interpretation of λ -reduction as a textual replacement results in the following expression, where the entire meaning representation of the noun phrase is embedded in the Server predicate.

$$\begin{aligned} &\exists e \text{ Isa}(e, \text{Serving}) \\ &\quad \wedge \text{Server}(e, \exists x \text{ Isa}(x, \text{Restaurant})) \wedge \text{Served}(e, \text{Meat}) \end{aligned}$$

Contd...



- The notion of a complex-term that allows FOPC expressions like $xIsa(x; Restaurant)$ to appear in places where normally only ordinary FOPC terms would appear.
- Formally, a complex-term is an expression with the following three-part structure.

< Quantifier variable body >

$$\exists e Isa(e, Serving) \\ \wedge Server(e, < \exists x Isa(x, Restaurant) >) \wedge Served(e, Meat)$$

Contd...



- As was the case with λ -expressions, this notational change will only be useful if we can provide a straightforward way to convert it into ordinary FOPC syntax.
- This can be accomplished by rewriting any predicate using a complex-term according to the following schema.

$P(< \textit{Quantifier variable body} >)$

$\Rightarrow$

$\textit{Quantifier variable body Connective } P(\textit{variable})$



Contd...

Complex-term:

1. Is extracted from the predicate in which it appears.
2. Is replaced by the variable that represents the object in question.
3. And has its variable, quantifier, body prepended to the new expression through the use of an appropriate connective.

The following pair of expressions illustrates this complex-term reduction on our current example

$$\begin{aligned} & \textit{Server}(e, < \exists x \textit{Isa}(x, \textit{Restaurant}) >) \\ \Rightarrow \\ & \exists x \textit{Isa}(x, \textit{Restaurant}) \wedge \textit{Server}(e, x) \end{aligned}$$

Contd...



The connective that is used to attach the extracted formula to the front of the new expression depends on the type of the quantifier being used: $\wedge$ is used with $\exists$, and $\Rightarrow$ is used with $\forall$.

- For example, if $A:sem$ is a complex-term then $A:sem:quantifier$, $A:sem:variable$, and $A:sem:body$ retrieve the complex-term's quantifier, variable, and body, respectively.
- Given the simple syntactic structure of this noun phrase, the job of the NP semantic attachment is fairly straightforward

$$NP \rightarrow Det\ Nominal \quad \{ < Det.sem\ x\ Nominal.sem(x) > \}$$

Contd...



- This attachment creates a complex-term consisting of a quantifier retrieved from the Det, followed by an arbitrary variable, and then an application of the λ -expression associated with the Nominal to that variable.
- This λ -application ensures that the correct variable appears within the predicate specified by the Nominal.
- The attachment for the determiner simply specifies the quantifier to be used.

$$Det \rightarrow a \quad \{\exists\}$$

Contd...



- The job of the nominal category is to create the Isa formula and λ expression needed for use in the noun phrase

$$\textit{Nominal} \rightarrow \textit{Noun} \quad \{\lambda x \textit{Isa}(x, \textit{Noun.sem})\}$$

- Finally, the noun attachment simply provides the name of the category being discussed.

$$\textit{Noun} \rightarrow \textit{restaurant} \quad \{\textit{Restaurant}\}$$

Contd...

This example, we have introduced five concrete mechanisms that instantiate the abstract functional characterization of semantic attachments that began this section.

1. The association of normal FOPC expressions with lexical items.
2. The association of function-like λ -expressions with lexical items.
3. The copying of semantic values from children to parents.
4. The function-like application of λ -expressions to the semantics of one or more children of a constituent.
5. The use of complex-terms to allow quantified expressions to be temporarily treated as terms.



Contd...

- The introduction of λ -expressions and complex-terms was motivated by the gap between the syntax of FOPC and the syntax of English.
- These extra-logical devices serve to bring the syntax of FOPC closer to the syntax of the language being processed thus facilitating the semantic analysis process.
- Meaning representations that make use of these kinds of devices are usually referred to as **quasi-logical forms or intermediate representations**.



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- The **quasi-logical form** is usually applied to representations that can easily be converted to a logical representation via some simple **syntactic transformation**.
- The **intermediate representation** is normally used to refer to meaning representations that serve as input to further analysis processes in an attempt to produce **deeper meaning representations**.

Quantifier Scoping and the Translation of Complex Terms

- To translate expressions containing complex terms into FOPC expressions is not unique.
- Consider the following example, along with its original unscoped meaning representation.

Every restaurant has a menu.

$$\begin{aligned} &\exists e Isa(e, Having) \\ &\quad \wedge Haver(e, < \forall x Isa(x, Restaurant) >) \\ &\quad \wedge Had(e, < \exists y Isa(y, Menu) >) \end{aligned}$$


Contd...



- If the complex-terms filling the *Haver* and the *Had* roles are rewritten so that the quantifier for the Haver role has the outer scope, then the result is the following meaning representation, results in common sense interpretation

$$\forall x \text{Restaurant}(x) \Rightarrow \\ \exists e, y \text{Having}(e) \wedge \text{Haver}(e, x) \wedge \text{Isa}(y, \text{Menu}) \wedge \text{Had}(e, y)$$

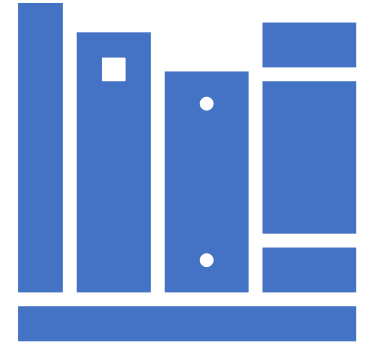
- On the other hand, if the terms are rewritten in the reverse order. The one menu all restaurant share

$$\exists y \text{Isa}(y, \text{Menu}) \wedge \forall x \text{Isa}(x, \text{Restaurant}) \Rightarrow \\ \exists e \text{Having}(e) \wedge \text{Haver}(e, x) \wedge \text{Had}(e, y)$$

Contd....



- Example illustrates the problem of ambiguous *quantifier scoping* a single logical formula with two complex terms gives rise to two distinct and incompatible FOPC representations.
- In the worst case, sentences with N quantifiers will have $O(N!)$ different possible quantifier scopings.
- **Domain specific knowledge** can then be used to either **accept** a quantified formula, or **reject it** and **request another formula**.



Module IV-Attachments for a Fragment of English

Introduction



- Describes a set of semantic attachments for a small fragment of English.
- Sentence

Sentence



- The sentences with imperatives, Yes/No questions, and WH questions. Let's start by considering the following examples.

Flight 487 serves lunch - convey factual information to a hearer

Serve lunch - is a request for an action

Does Flight 207 serve lunch? are requests for information

Which flights serve lunch? - are requests for information

- The meaning representations of these examples all contain propositions concerning the serving of lunch on flights.
- They differ with respect to the role that these propositions are intended to serve.

Contd...

- To capture these differences - a set of operators is applied to FOPC sentences.
- Specifically, the following operators will be applied to the FOPC representations
 - DCL - declaratives
 - IMP - imperatives
 - YNQ - yes-no questions
 - WHQ - wh-question



Contd...



$$S \rightarrow NP VP \quad \{DCL(VP.sem(NP.sem))\}$$

- The normal interpretation for a representation headed by the DCL operator would be as a factual statement to be added to the current knowledge-base.
- Imperative sentences begin with a verb phrase and lack an overt subject. Because of the missing subject, the meaning representation for the main verb phrase will consist of a λ -expression with an unbound λ -variable representing this missing subject.

Contd...

- Simply supply a subject to the λ -expression by applying a final λ -reduction to a dummy constant.
- The IMP operator can then be applied to this representation as in the following semantic attachment.

$$S \rightarrow VP \quad \{IMP(VP.sem(DummyYou))\}$$

- Applying this rule

$$IMP(\exists e Serving(e) \wedge Server(e, DummyYou) \wedge Served(e, Lunch))$$

- Imperatives can be viewed as a kind of speech act



Contd...

- yes-no-questions consist of a sentence initial auxiliary verb, followed by a subject noun phrase and then a verb phrase.
- The following semantic attachment simply ignores the auxiliary, and with the exception of the YNQ operator

$$S \rightarrow Aux NP VP \quad \{YNQ(VP.sem(NP.sem))\}$$

$$YNQ(\exists e Serving(e) \wedge Server(e, Flt207) \wedge Served(e, Lunch))$$

- Yes or No Questions should be thought as asking the whether the propositional part of its meaning is true or false given the knowledge currently contained in the knowledge-base.



Contd...

- wh-subject-questions ask for specific information about the subject of the sentence rather than the sentence as a whole.
- The following attachment produces a representation that consists of the operator WHQ, the variable corresponding to the subject of the sentence, and the body of the proposition.

$$S \rightarrow WhWord NP VP \quad \{WHQ(NP.sem.var, VP.sem(NP.sem))\}$$

$$WHQ(x, \exists e, x Isa(e, Serving) \wedge Server(e, x) \\ \wedge Served(e, Lunch) \wedge Isa(x, Flight))$$



Contd...

- Such questions can be answered by returning a set of assignments for the subject variable that make the resulting proposition true with respect to the current knowledge-base.
- Finally, consider the following wh-non-subject-question.

How can I go from Minneapolis to Long Beach?

- The question is not about the subject of the sentence but rather some other argument, or some aspect of the proposition as a whole.
- In this case, the representation needs to provide an indication as to what the question is about.



Contd...

- The following attachment provides this information by providing the semantics of the auxiliary as an argument to the WHQ operator.

$$S \rightarrow WhWord Aux NP VP \quad \{WHQ WhWord.sem VP.sem(NP.sem)\}$$

- Rule:

$$WHQ(How, \exists e Isa(e, Going) \wedge Goer(e, User) \\ \wedge Origin(e, Minn) \wedge Destination(e, LongBeach))$$


Noun phrase



- The meaning representations for noun phrases can be either normal FOPC terms or complex-terms.
- the semantic attachments needed to produce meaning representations for some of the most frequent kinds of English noun phrases
- Compound nominals, also known as noun-noun sequences, consist of simple sequences of nouns.

Flight schedule

Summer flight schedule



Contd...

- CFG Rule for Noun

Nominal $\rightarrow$ *Noun*

Nominal $\rightarrow$ *Noun Nominal*

- The relation NN is used to specify that a relation holds between the modifying elements of a compound nominal and the head Noun

Nominal $\rightarrow$ *Noun Nominal*

$\{\lambda x \text{ Nominal.sem}(x) \wedge \text{NN}(\text{Noun.sem}, x)\}$

Contd...



- This representation correctly instantiates a term representing a *Schedule*, while avoiding the creation of terms representing either a *Flight* or *Summer*.

$$\lambda x Isa(x, Schedule) \wedge NN(x, Flight)$$

$$\lambda x Isa(x, Schedule) \wedge NN(x, Flight) \wedge NN(x, Summer)$$

Genitive Noun phrases



- **Genitive noun phrases** make use of complex determiners that consist of noun phrases with possessive markers, as in
 - ✓ Atlanta's airport
 - ✓ Maharani's menu.

Contd...



- It is quite tempting to represent the relation between these words as an abstract kind of possession.
- A little introspection, however, reveals that the relation between a city and its airport has little in common with a restaurant and its menu.

$$NP \rightarrow \textit{ComplexDet Nominal} \\ \{ \langle \exists x \textit{Nominal.sem}(x) \wedge GN(x, \textit{ComplexDet.sem}) \rangle \}$$

$$\textit{ComplexDet} \rightarrow NP \text{ 's} \quad \{NP.sem\}$$

Contd...



- Applying these rules to Atlanta's airport results in the following complex term

$$< \exists x Isa(x, Airport) \wedge GN(x, Atlanta) >$$

- The semantic interpretation would have to determine that the relation denoted by the relation GN is actually a location.

Adjective Phrases

- English adjectives can be split into two major categories: pre-nominal and predicate.
- These categories are exemplified by the following BERP examples.

I don't mind a cheap restaurant.

This restaurant is cheap.

- For the pre-nominal case, an obvious and often incorrect proposal for the semantic attachment is illustrated in the following rules.

$$\begin{aligned} & \textit{Nominal} \rightarrow \textit{Adj Nominal} \\ & \quad \{\lambda x \textit{Nominal.sem}(x) \wedge \textit{Isa}(x, \textit{Adj.sem})\} \\ & \textit{Adj} \rightarrow \textit{cheap} \quad \{\textit{Cheap}\} \end{aligned}$$


Contd...



- This solution modifies the semantics of the nominal by applying the predicate provided by the adjective to the variable representing the nominal. For our cheap restaurant example, this yields the following fairly reasonable representation.

$$\lambda x \text{ } Isa(x, \textit{Restaurant}) \wedge Isa(x, \textit{Cheap})$$

- In this case, this amounts to the intersection of the category of cheap things with the category of restaurants. This solution often does the wrong thing.

Contd...

- For example, consider the following meaning representations for the phrases small elephant, former friend, and fake gun.

$$\lambda x \text{ Isa}(x, \textit{Elephant}) \wedge \text{Isa}(x, \textit{Small})$$
$$\lambda x \text{ Isa}(x, \textit{Friend}) \wedge \text{Isa}(x, \textit{Former})$$
$$\lambda x \text{ Isa}(x, \textit{Gun}) \wedge \text{Isa}(x, \textit{Fake})$$


- The first - particular elephant is a member of the general category of small things, which is probably not true.
- The second example is strange in two ways: it asserts that the person in question is a friend, which is false, and it makes use of a fairly unreasonable category of former things.

Contd...

- The third example asserts that the object in question is a gun despite the fact that fake means it is not one.
- The best approach is to simply note the status of a specific kind of modification relation.
- Further procedure with access to additional relevant knowledge can replace this vague relation with an appropriate representation.

$$\begin{aligned} & \textit{Nominal} \rightarrow \textit{Adj Nominal} \\ & \{\lambda x \textit{Nominal.sem}(x) \wedge \textit{AM}(x, \textit{Adj.sem})\} \end{aligned}$$



- Applying this rule to a cheap restaurant results in the following formula

$$\exists x \textit{Isa}(x, \textit{Restaurant}) \wedge \textit{AM}(x, \textit{Cheap})$$

Verb Phrases

- The general schema for computing the semantics of verb phrases relies on the notion of function application.
- The λ -expression attached to the verb is simply applied to the semantic attachments of the verb's arguments.

Infinitive Verb Phrases:

- A fair number of English verbs take some form of verb phrase as one of their arguments.
- This complicates the normal verb phrase semantic schema since these argument verb phrases interact with the other arguments of the head verb.

Contd...

- Consider the following example.

I told Harry to go to Maharani.

- The meaning representation for this example should be something like the following

$$\begin{aligned} \exists e, f, x \text{ } & Isa(e, Telling) \wedge Isa(f, Going) \\ & \wedge Teller(e, Speaker) \wedge Tellee(e, Harry) \wedge ToldThing(e, f) \\ & \wedge Goer(f, Harry) \wedge Destination(f, x) \end{aligned}$$

- The first is that it consists of two events.
- The second is that one of the participants, Harry, plays a role in both of the two events.

Contd...

- Semantically, we can interpret this subcategorization frame for Tell as providing three semantic roles: a person doing the telling, a recipient of the telling, and the proposition being conveyed.

$$\lambda x, y \lambda z \exists e \text{ } Isa(e, Telling) \\ \wedge Teller(e, z) \wedge Tellee(e, x) \wedge ToldThing(e, y)$$

- *Harry* plays the role of both the *Tellee* of the *Telling* event and the *Goer* of the *Going event*.
- Harry is not available when the Going event is created within the infinitive verb phrase.

Contd...

- This problem will be solved by
- simply applying the semantics of the verb to the semantics of the other arguments of the verb as follows

$$VP \rightarrow \textit{Verb NP VPto} \quad \{ \textit{Verb.sem}(\textit{NP.sem}, \textit{VPto.sem}) \}$$

- Since the **to** in the infinitive verb phrase construction does not contribute to its meaning, we simply copy the meaning of the child verb phrase up to the infinitive verb phrase.

Contd...

$$VPto \rightarrow to VP \quad \{VP.sem\}$$

- In this solution, the verb's semantic attachment has two tasks: incorporating the NP.sem, the Goer, into the VPto.sem, and incorporating the Going event as the ToldThing of the Telling. The following attachment performs both tasks.

$$\begin{aligned} Verb &\rightarrow tell \\ &\{ \lambda x, y \\ &\quad \lambda z \\ &\quad \exists e, y. variable\ Isa(e, Telling) \\ &\quad \quad \wedge Teller(e, z) \wedge Tellee(e, x) \\ &\quad \quad \wedge ToldThing(e, y.variable) \wedge y(x) \end{aligned}$$

Contd...

- In this approach, the λ -variable x plays the role of the **Tellee of the telling** and the **argument to the semantics of the infinitive**
- The expression $y(x)$ represents a **λ -reduction** that inserts Harry into the Going event as the Goer.
- The notation $y:\text{variable}$, is analogous to the notation used for complex-term variables, and gives us access to the event variable representing the Going event within the infinitive's meaning representation.

Prepositional Phrases

- Prepositional phrases serve two distinct functions:
 - They assert binary relations between their heads and the constituents to which they are attached.
 - They signal arguments to constituents that have an argument structure.
- These two functions argue for two distinct types of prepositional phrases that differ based on their semantic attachments.
- consider three places in the grammar where prepositional phrases serve these roles: **modifiers of noun phrases**, **modifiers of verb phrases**, and **arguments to verb phrases**.

Nominal Modifier Prepositional Phrases

- Modifier prepositional phrases denote a binary relation between the concept being modified, which is external to the prepositional phrase, and the head of the prepositional phrase.
- Consider the example and its meaning representation.

A restaurant on Pearl

$\exists x \text{ Isa}(x, \text{Restaurant}) \wedge \text{On}(x, \text{Pearl})$

The relevant grammar rules:

$NP \rightarrow Det \text{ Nominal}$

$Nominal \rightarrow Nominal \text{ PP}$

$PP \rightarrow P \text{ NP}$

Nominal Modifier Prepositional Phrases

- Proceeding in a bottom-up fashion, the semantic attachment for this kind of relational preposition should provide a two-place predicate with its arguments distributed over two λ -expressions.

$$P \rightarrow on \quad \{\lambda y \lambda x \text{ On}(x, y)\}$$

- The first argument to the predicate is provided by the head of prepositional phrase and the second is provided by the constituent that the prepositional phrase is ultimately attached to. The following semantic attachment provides the first part.

$$PP \rightarrow P NP \quad \{P.sem(NP.sem)\}$$

Contd...

- This λ -application results in a new λ -expression where the remaining argument is the inner λ -variable. This remaining argument can be incorporated using the following nominal construction.

$$\textit{Nominal} \rightarrow \textit{Nominal PP} \quad \{\lambda z \textit{Nominal.sem}(z) \wedge \textit{PP.sem}(z)\}$$

Verb Phrase Modifier Prepositional Phrases

- The general approach to modifying verb phrases is similar to that of modifying nominals.
- The differences lie in the details of the modification in the verb phrase rule; the attachments for the preposition and prepositional phrase rules are unchanged.
- Let's consider the phrase ate dinner in a hurry which is governed by the following verb phrase rule.

$$VP \rightarrow VP PP$$

- The meaning representation of the verb phrase constituent in this construction, ate dinner, is a λ -expression where the λ variable represents the as yet unseen subject.

Contd...

- The representation of the prepositional phrase is also a λ -expression where the λ variable is the second argument in the PP semantics.

$$\lambda x \exists e \text{ Isa}(e, \text{Eating}) \wedge \text{Eater}(e, x) \wedge \text{Eaten}(e, \text{Dinner})$$

- The correct representation for the modified verb phrase should contain the conjunction of these two representations with the Eating event variable filling the first argument slot of the In expression.

$$\lambda x \text{ In}(x, < \exists h \text{ Hurry}(h) >)$$

- In addition, this modified representation must remain a λ -expression with the unbound Eater variable as the new λ -variable.
- The following attachment expression fulfills all of these requirements

$$VP \rightarrow VP PP \quad \{\lambda y VP.sem(y) \wedge PP.sem(VP.sem.variable)\}$$

- There are two aspects of this attachment that require some elaboration.
- The first involves the application of the constituent verb phrases' λ expression to the variable y .
- Binding the lower λ -expression's variable to a new variable allows us to lift the lower variable to the level of the newly created λ -expression.
- The result of this technique is a new λ -expression with a variable.
- This allows a λ -expression to be modified during the analysis process before the argument to the expression is actually available

Contd...

- The second new aspect in this attachment involves the *VP:sem:variable* notation.
- This notation is used to access the event-variable representing the underlying meaning of the verb phrase, in this case, *e*.
- This is analogous to the notation used to provide access the various parts of complex-terms introduced earlier.
- Applying this attachment to the current example yields the following representation, which is suitable for combination with a subsequent subject noun phrase.

$$\lambda y \exists e \text{ Isa}(e, \text{Eating}) \wedge \text{Eater}(e, y) \wedge \text{Eaten}(e, \text{Dinner}) \\ \wedge \text{In}(e, < \exists h \text{Hurry}(h) >)$$

Verb Argument Prepositional Phrases

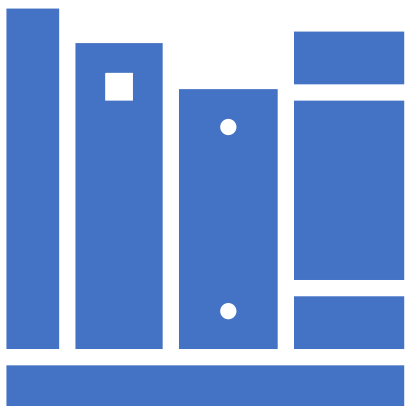
- The prepositional phrases in this category serve to signal the role an argument plays in some larger event structure.
- As such, the preposition itself does not actually modify the meaning of the noun phrase.
- Consider the following example of role signaling prepositional phrases.

I need to go from Boston to Dallas.

- The arguments to go are expressed as a prepositional phrase.
- However, the meaning representations of these phrases should consist solely of the unaltered representation of their head nouns.

$$PP \rightarrow P NP \quad \{NP.sem\}$$

- https://www.youtube.com/watch?v=YVC-L_ILQb4



Thank you
